



Universidad Tecnológica de Bolívar

Assignment 5 (05/16/2026)

Linear Algebra and Least Squares Regression

Student:

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The purpose of these problems is to use numerical approaches for determining the solutions to linear systems of equations and data fitting by the method of least squares.

Problem 1

This is an example problem taken from [Kong Q. and M. \(2020\)](#):

Write a function `my_rec_det(M)` where the output is $\det(M)$. Use cofactors to compute the determinant, not the NumPy's function. Use the fact that for an $n \times n$ square matrix M we have

$$\det(M) = \sum_{i=1}^n (-1)^{i-1} M(1, i) \det(m_{1,i})$$

where $m_{1,i}$ is the minor of M created by removing the 1st row and the i th column.

Problem 2

This is an example problem taken from [Kong Q. and M. \(2020\)](#):

Consider the following network shown in Fig.1 consisting of two power supply stations denoted by S1 and S2 and five power recipient nodes denoted by N1 to N5. The nodes are connected by power lines, which are denoted by arrows, and power can flow between nodes along these lines in both directions.

Let d_i be a positive scalar denoting the power demand for node i ; assume that this demand must be met exactly. The capacity of the power supply stations is denoted by S . Power supply stations must run at capacity. For each arrow, let f_j be the power flow along that arrow. Negative flow implies that power is running in the opposite direction of the arrow.

Write a function `my_flow_calculator(S,d)` where S is a 1×2 vector representing the capacity of each power supply station, and d is a 1×5 row vector representing the demands at each node (i.e., $d[0]$ is the demand at node 1). The output argument, f , should be a 1×7 row vector denoting the flows in the network (i.e., $f[0] = f_1$ in the diagram). The flows contained in f should satisfy all the constraints of the system, like power generation and demands. Note that there may be more than one solution to the system of equations. The total flow into a node must equal the total flow out of the node plus the demand; that is, for each node i , $f_{inflow} = f_{outflow} + d_i$. You may assume that $\sum S_j = \sum d_i$.

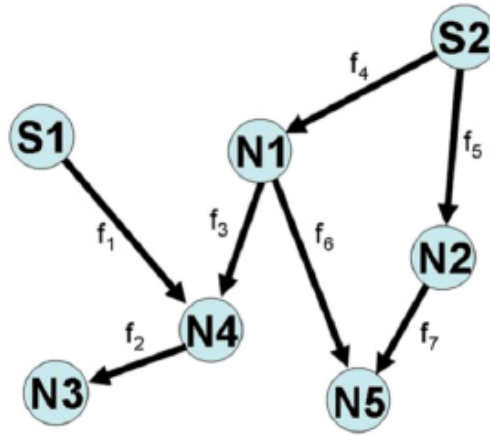


Figure 1: Graph for Problem 2.

Problem 3

This is an example problem taken from [Kreyszig \(2006\)](#):

Write a program for the Gauss elimination with pivoting. Apply it to the following problems:

$$\begin{array}{lcl}
 & 3x_2 & +5x_3 = 0 \\
 \text{a)} \quad & 3x_1 & -4x_2 = 0 \\
 & 5x_1 & +6x_3 = 0 \\
 \\
 & 6.4x_1 & +3.2x_2 = -1.6 \\
 \text{b)} \quad & 3.2x_1 & -1.6x_2 +4.8x_3 = 32.0 \\
 & & +4.8x_2 -9.6x_3 +7.2x_4 = -78.0 \\
 & & & 7.2x_3 +4.8x_4 = 20.4
 \end{array}$$

Experiment with systems whose coefficient determinant is small in absolute value. Also investigate the performance of your program for larger systems of your choice, including sparse systems.

Problem 4

This is an example problem taken from [Kreyszig \(2006\)](#):

Apply the Gauss-Seidel iteration (3 steps) to the system

$$\begin{array}{rrcr} 10x_1 & +x_2 & +x_3 & = 6 \\ x_1 & +10x_2 & +x_3 & = 6 \\ x_1 & +x_2 & +10x_3 & = 6 \end{array}$$

starting from

a) 0, 0, 0

b) 10, 10, 10

Compare and comment.

Problem 5

This is an example problem taken from [Kreyszig \(2006\)](#):

Estimate the resistance R from the least squares line that fits $(i, U) = (2.0, 104), (4.0, 206), (6.0, 314), (10.0, 530)$.

Problem 6

This is an example problem taken from [Kreyszig \(2006\)](#):

Estimate the average speed v_{av} of a car traveling according to $s = v \cdot t$ [km] (s =distance traveled, t [h]=time) from $(t, s) = (9, 140), (10, 220), (11, 310), (12, 410)$.

Problem 7

This is an example problem taken from [Kreyszig \(2006\)](#):

Estimate the spring modulus k from the force F [lbf] and extension s [cm], where $(F, s) = (1, 0.50), (2, 1.02), (4, 1.99), (6, 3.01), (10, 4.98), (20, 10.03)$.

Problem 8

This is an example problem taken from [Kreyszig \(2006\)](#):

Fit a parabola to a worker's time on duty and reaction time given by the points (x, y) using least squares

$$\begin{array}{c|ccccc} x \text{ [h]} & 1 & 2 & 3 & 4 & 5 \\ y \text{ [seg]} & 1.50 & 1.28 & 1.40 & 1.85 & 2.20 \end{array}$$

where x is the time on duty and y is the reaction time.

Problem 9

This is an example problem taken from [Kong Q. and M. \(2020\)](#):

Write a function `my_lin_regression(f, x, y)` where `f` is a list containing function objects to basis functions, and `x` and `y` are arrays containing noisy data. Assume that `x` and `y` are the same size. Let an estimation function for the data contained in `x` and `y` be defined as $\hat{y}(x) = \beta(1) \cdot f_1(x) + \beta(2) \cdot f_2(x) + \dots + \beta(n) \cdot f_n(x)$, where `n` is the length of `f`. Your function should compute `beta` according to the least squares regression formula.

Test case: Note that your solution may vary by a little bit, depending on the random numbers generated.

```
x = np.linspace(0, 2*np.pi, 1000)
y = 3*np.sin(x) - 2*np.cos(x) + np.random.random(len(x))
f = [np.sin, np.cos]
beta = my_lin_regression(f, x, y)
plt.figure(figsize = (10,8))
plt.plot(x,y,"b.", label = "data")
plt.plot(x, beta[0]*f[0](x)+beta[1]*f[1](x)+beta[2], "r", label="regression")
plt.xlabel("x")
plt.ylabel("y")
plt.title("Least Square Regression Example")
plt.legend()
plt.show()
```

References

Kong Q., S. T. and M., B. A. (2020). *Python Programming and Numerical Methods: A Guide for Engineers and Scientists*. Elsevier.

Kreyszig, E. (2006). *Advanced Engineering Mathematics*. John Wiley & Sons.